# N-Queens

Interactive backtracking demo for svg-dom-graph | Design proposal, Revision 6 | 3 November 2026

**Status:** Behavioural specification ready for Phase 0 repository/API verification. Revision 6 supersedes Revision 5 and restores explicit regression coverage, fixes seek/change ordering, and makes focus and pacing rules testable.

## Revision 6 decisions

| Area | Binding decision |
|---|---|
| Seek completion | A range change flushes any pending seek reconciliation synchronously before announcing; on failure announce the error, not an uncommitted index. |
| Focus visibility | Custom topmost ring appears only while focus is within the board grid, follows keyboard movement and pointer click, hides on grid blur, and remains visible in forced-colors. |
| Play | Projection commits synchronously. Only seek input may await the next animation frame; Play never queues a new tick until its preceding reconcile returns successfully. |
| Verification | Full V4/V5 trace, visual, status, boundary focus, strict-jump and resource-usage tests are mandatory. |
| Unchanged | N=4..9 interactive; compact snapshots; deterministic event budget and solver; ray/queen layering; dirty repair before boundary no-ops. |

## 1. Purpose and scope

Visualize every deterministic depth-first N-Queens decision on an inspect-only, accessible SVG board. Exercise random-access trace projection, fixed overlay geometry, layer ordering, transactional DOM error handling, focus retention, playback scheduling and CDP verification. Phase 0 must verify existing repository interfaces; no public crate API expansion is assumed.

Interactive selector: N=4..9 (default 8). Materializing trace builder: N=1..9. Event budget: 250,000 events. N=10 is count-only and cannot be selected or materialized.

## 2. Coordinates and board

Internal rows and columns are zero-based, increasing downward/rightward; positions[r] = c. Rust fixtures use zero-based vectors, e.g. [1,3,0,2]. UI status and solution vectors use one-based coordinates, e.g. [2,4,1,3], explicitly labelled “columns by row (1-based)”. SVG IDs may use zero-based indices. Board interaction changes focus only, never the search.

## 3. Deterministic solver and event stream

Rows and columns are tried in ascending order without symmetry pruning. Conflicts are an earlier queen in the same column, main diagonal (r-c), or anti-diagonal (r+c). Reported reason precedence: Column, MainDiagonal, AntiDiagonal. K counts distinct conflicting directions (not queens). Every successful placement is removed exactly once.

| Event | Transition and required presentation |
|---|---|
| Try {r,c} | Committed positions unchanged; candidate outlined. Draw three complete clipped rays (column, both diagonals) identically whether legal or illegal; no legality hint. |
| Reject {r,c,reason,K} | Committed positions unchanged; rejected pattern. Exactly the precedence-selected ray is thick and solid; the other two are thin and dashed regardless of whether they also conflict. Show “reason” if K=1; “reason (first of K conflicting directions)” if K>1. |
| Place {r,c} | Commit queen; remove candidate and rays; update queen glyph, outline and status. |
| Solution | Immediately follows last-row Place; keep complete queens, show solution frame and ordinal. No duplicated positions event payload. |
| Remove {r,c} | Undo latest queen; recently removed outline only, no rays. |

| Event | Transition and required presentation |
|---|---|
| Exhausted | Emit exactly once after traversal; empty board and no transients; Play stops. |

Snapshot 0 is empty; event k produces snapshot k. M events imply M+1 snapshots. Every Try is followed immediately by Reject or Place. Completion order: Place, Solution, Remove. Next at M changes no index.

## 4. Counts, validation and storage

| N | Events | Solutions | Mode |
|---|---|---|---|
| 4 | 139 | 2 | Interactive |
| 8 | 33,589 | 92 | Interactive |
| 9 | 153,502 | 352 | Interactive |
| 10 | 732,563 | 724 | Count-only |

The builder rejects N=0 as InvalidBoardSize and N>9 as UnsupportedBoardSize **before** indexing, allocating or generating events. The count-only traversal accepts N=1..10, rejecting 0 and >10 analogously. Event 250,001 raises EventBudgetExceeded; never return a partial successful trace or fabricate Exhausted.

Use a flat Vec<CompactSnapshot>, each holding positions: [u8;9] and depth: u8. 0xFF is the only unplaced-row sentinel; depth is a count 0..N. For r<depth, positions[r]<N; for r>=depth, positions[r]=0xFF. Keep immutable SearchEvent metadata and sorted solution_indices; binary-search strict solution jumps. No per-snapshot allocation and no duplicated solution-position payload.

Count-only traversal follows the same event policy but omits snapshots and materialization limits. It is a cross-check, **not** an independent oracle: hard-coded expected totals and solutions are the reference. Measure size_of, Vec capacity, estimated N=9 memory, build/seek latency.

## 5. Projection, layering and error recovery

### 5.1 One attack model and exact layer order

Exactly three clipped geometric ray paths exist in a noninteractive overlay. No per-cell attacked state, tint or attacked ARIA label. Try rays do not depend on queen occupancy. On Reject, the chosen direction is thick/solid; other rays thin/dashed. Ray geometry is fixed and at most three paths are visible.

| Back-to-front layer | Visual / constraints |
|---|---|
| 1. Cell fills | Persistent checkerboard; no state-dependent $N^2$ repaint. |
| 2. Ray overlay | At most three paths, pointer-events:none and aria-hidden; above fills, below queens. |
| 3. Queen glyphs | Persistent independent glyph layer, shown/hidden from committed state. |
| 4. Transient outlines | Candidate, rejected or recently removed; mutually exclusive and above rays. |
| 5. Solution frame | Board-wide only on Solution snapshots; no overlap with transient outlines. |
| 6. Focus ring | Topmost custom offset dual-stroke/non-colour ring; remains distinct over candidate/rejected/removed outlines. |

Each square remains one semantic gridcell hit/focus target, while fill, glyph and transient outline have separately controlled SVG visual ownership. Phase 0 verifies that Scene/SVG z-order supports these layers without duplicating semantic cells; otherwise document and approve a spec amendment **before Phase 2**. Never silently collapse layers or place rays above glyphs.

### 5.2 Focus-ring visibility, including high contrast

Show the custom ring exactly while keyboard or pointer focus is within the grid, on its current roving cell. It follows arrow/Home/End navigation and pointer click, including mouse click (do not rely solely on :focus-visible). Hide when focus leaves the grid. In forced-colors mode use system-colour strokes (e.g. Highlight and Canvas), a visible offset/gap and contrasting stroke widths/dash where needed; do not rely on colours alone or disable the ring. Focus is preserved across snapshot changes, clamped/re-homed on an N rebuild.

### 5.3 Transactional projection and dirty recovery

Commit the canonical selected index only after synchronous DOM reconciliation succeeds. Retain last canonical state and dirty flag. On partial write failure: set dirty, cancel Play, report error, keep the canonical index. Every explicit navigation action (Prev/Next, Restart, seek, solution jump or Play) first attempts a full redraw of the canonical state if dirty. Only after repair succeeds may the action change indices or start Play. Failed repair leaves dirty and keeps the index.

Repair-before-no-op includes Prev at 0, Next at M, absent solution jump, same-index seek, Play at M and aria-disabled boundary controls, whose handlers remain callable. A clean boundary action never changes index; Play at M never schedules. Rendering errors while selecting a new target preserve canonical index and mark the view dirty. Teardown removes listeners, callbacks and observers.

## 6. Status, legend, navigation and timing

### 6.1 Required status fields

| Field | Display contract |
|---|---|
| Position | Step i/M (initial step 0). |
| Event kind | Initial, Try, Reject, Place, Solution, Remove, Exhausted. |
| Active cell | Row and column, one-based where present; otherwise dash. |
| Depth | Committed queen count / N. |
| Reject reason | Chosen column/main/anti-diagonal. Append “first of K conflicting directions” only for K>1. |
| Solutions | Count found so far / known total; Solution shows ordinal. |
| Solution vector | On Solution show one-based columns by row; otherwise hidden/dash. |
| Playback | Playing or Paused; selected 1/5/20/50 events per second. |

### 6.2 Mandatory legend

Placed queen; Candidate (Try); Rejected candidate; Recently removed queen; Candidate rays (all three directions); Chosen conflict ray (Reject, thick/solid); Solution frame; Keyboard focus ring. Each key must be distinguishable by geometry, dash, stroke or pattern as well as colour. Show dimension and one-based row/column orientation.

### 6.3 Controls, seeking and boundaries

| Control | Semantics |
|---|---|
| Prev / Next / Restart | Single-step within [0,M]; Restart stops Play and goes to 0. Repair dirty view first even for boundary no-op. |
| Previous / next Solution | Select greatest solution index strictly less than i / smallest strictly greater than i. No matching index is a no-op after dirty repair; never land on current Solution. |
| Seek range | Native labelled range, min=0 max=M step=1. On input, latest requested index wins; coalesce projection to at most one animation frame. No intermediate trace replay. |
| Seek change | Synchronously flush pending latest requested reconciliation before announcing the final **committed** index once. Cancel pending frame to avoid duplicate render. If flush fails, announce error, not uncommitted index; canonical index unchanged/dirty. |
| Precision | Range Left/Right move exactly one index; PageUp/PageDown may use native larger increments and are tested as such. Prev/Next are exact fallback. N=9 pointer positioning is intentionally coarse. |
| Boundaries | aria-disabled rather than native disabled, retains control focus and handler for dirty repair. |
| Play / Pause | 1,5,20,50 events/s, one loop. Play at M is no-op after possible repair. Manual step, seek, jump, restart, N change, error or teardown cancels Play. |
| N selector | Only N=4..9; cancel playback first, rebuild, and clamp/rehome roving board focus. |

The range thumb/native announcement may temporarily indicate the *requested* index before the next frame; visible board and status always describe the **last committed** index and are authoritative. This short lag is accepted. No polite-region message on input. On change, flush before the one status announcement. Invalidate stale callbacks on newer request, N change, teardown or a synchronous flush.

### 6.4 Playback scheduler

Use one replaceable demo-local scheduler, compatible with the existing CDP time-control harness. Playback DOM reconciliation and index commit are **synchronous**; no playback reconcile is deferred. At most one scheduled tick exists. On each tick: compute next index, synchronously reconcile, commit on success, **then** schedule the next interval. Thus pending Play reconciliation cannot overlap by construction; only *seek input* may have a frame-deferred projection. Slow frames create no backlog or catch-up burst, so achieved speed may be below nominal.

Use monotonic/fixed-delay timing with injectable deterministic ticks; generation/epoch invalidation discards callbacks after Pause, failure, N change, rebuild and teardown. No tick after Exhausted/M. Phase 0 checks actual requestAnimationFrame/timeout behaviour in CDP and adopts a scheduler controlled by its test seam.

## 7. Board semantics and screen-reader policy

Use role=grid, role=row and role=gridcell, or a verified equivalent using existing svg-dom-graph accessibility support. Structural grid context supplies row/column, so cell accessible names describe semantic state only: empty, queen, candidate, rejected with chosen reason, removed. Exactly one cell has tabindex=0, all others -1.

Arrows navigate without wrap; Home/End first/last in row; Ctrl+Home/End board corners. Board PageUp/PageDown are not intercepted. Enter/Space never place queens. Pointer click transfers roving focus to the clicked cell, showing the custom ring, but never changes search state or event index. Ray overlay has no hit target.

Paused/manual navigation announces one concise step summary. During Play suppress Try/Reject/Place/Remove announcements; announce only Solution and Exhausted. Pause announces index, depth and solutions found. Native slider value may precede committed state while dragging, but the status remains authoritative; the polite region announces once after successful change flush, or announces the failure.

## 8. Deterministic test matrix - required coverage

| Test group | Mandatory evidence / assertions |
|---|---|
| Solver and literals | Exact N=4 solutions [1,3,0,2], [2,0,3,1]; solution totals N=1..9: 1,0,0,2,10,4,40,92,352; event totals N=4/8/9/10: 139/33,589/153,502/732,563. |
| Bounds and limits | Materializer N=0 and 10 fail before indexing/allocation; count-only N=0 and 11 reject, N=10 produces 724 solutions; attempted event 250,001 fails with no Exhausted. |
| All snapshots / Rust | Exhaustively traverse N=8 and N=9 snapshot arrays; depth/sentinel/queen positions, Try pairing, Place-Solution-Remove order, final empty board, ordered solution indices. |
| Try ray invariance | CDP and pure presentation-model assertions: three clipped rays and exact geometry/style are identical for legal and illegal Try at same candidate coordinates; no occupancy-dependent hint. |
| Reject and precedence | Recompute column/main/anti conflicts; K equals number of conflicting directions; selected reason follows column then main then anti priority. Chosen ray thick/solid and remaining two thin/dashed, including when other rays are also conflicting. K=1 has no “first of 1” suffix. |
| Status and legend | Check each exact field from 6.1 across Initial/Try/Reject/Solution/Exhausted, one-based coordinates, solution ordinal/vector and playback; all eight 6.2 legend entries and non-colour cues. |
| Strict solution jumps | From arbitrary index and from a Solution index, previous result is strictly smaller, next strictly greater. Missing result no-op; never reselect same Solution. |
| Dirty no-op boundaries | Inject partial DOM failure at M then Next; at 0 then Prev; absent solution jump; same-index seek; Play at M. Confirm repair first, unchanged index, error retained on failed repair. |
| Boundary keyboard focus | aria-disabled Prev/Next/jump remains focusable at boundary; handler repairs dirty view without focus loss; tabindex/roving target preserved. |
| CDP seek and change | Direct jumps to 0,1,20,000,M and Solution index; sampled event kinds. Many input events produce at most one render/frame. change before frame synchronously flushes latest requested state then announces exactly once; injected flush failure announces error. Keyboard arrow precision, native Page keys and stale callback cancellation. |
| Playback timing | Manual/injected scheduler reaches Exhausted, stops at M; commit-before-reschedule, slow render no backlog, at most one scheduled tick; cancel stale callbacks on pause/error/rebuild/teardown. |
| Visual and screen reader | CDP light/dark and forced-colors snapshots: rays over fills/under queens, rays over candidate, focus ring over Try/Reject/Remove. Grid roles and roving navigation, click-to-focus, focus hides on grid blur, Play announcements suppressed. |
| Resource/lifetime | Stable semantic cell count and persistent geometry over all sampled steps; no per-cell attacked mutations; <=3 ray paths; bounded timer count (<=1 Play and <=1 pending seek frame), no retained listeners/observers after teardown; N=9 memory/build/seek results. |

## 9. Delivery phases and acceptance gates

| Phase | Exit criteria |
|---|---|
| 0. Repository API | Verify Scene ownership and layer z-order, overlay paths, styling and toolbar callbacks, semantic grid, root focus preservation, CDP virtual time. Prototype split layering, boundary dirty repair and scheduler. Document signatures and necessary amendments. |
| 1. Solver/storage | Range and budget validation, exact fixtures and fixed event literals, compact [u8;9]+depth representation, sorted solution indices, exhaustive Rust snapshot tests. |
| 2. Board/projection | Separate visual layers, three rays, semantic grid, complete status and legend, sparse updates, synchronous transactional rendering, dirty repair, custom focus ring including forced-colors. |
| 3. Interaction/play | Strict solution navigation, seek input coalescing and synchronous change-flush, focus-safe boundaries, click-to-focus, replaceable fixed-delay scheduler and clean teardown. |
| 4. CDP/measurement | Full Section 8 matrix; browser regression screenshots and a11y; manual tick advancement; DOM failure/recovery; status/legend and resource assertions; N=9 memory and timings; demo docs/registration. |

## 10. Definition of done

For N=8/9 a user can select any snapshot, inspect all solutions, step or Play/Pause at 1/5/20/50 events/s without backlog. Try never reveals legality. Reject gives the precedence-selected reason, correct K and non-colour ray emphasis. The status and legend match Sections 6.1-6.2. A semantic grid offers accessible roving focus, visible custom ring including forced-colors, pointer focus without changing the algorithm, and focus-safe boundary actions.

All explicit navigation attempts dirty repair, including no-op boundaries and Play at M. Seek change cannot announce an uncommitted index; failure announces error. Full Rust trace validation and sampled CDP verification cover every row in Section 8; timers and DOM observers are reclaimed; no unbounded work, attacked-cell repaint or duplicated Play loops. Phase 0 repository verification remains the sole gate; compositing/API limitations require a documented spec amendment.

## Revision history

V1: initial concept. V2: event budgets and controls. V3: attack rays, compact snapshots and virtual time. V4: random access, status/legend and focus. V5: dirty no-op repair, split layers and playback pacing. **V6:** restored complete regression matrix; synchronous seek-change flush; explicit custom focus visibility/forced-colors; K=1 wording; synchronous Play versus deferred seek, and accepted native-range/status lag.